Diag Solutions

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1. **1.** The angle of elevation of the top of a chimney from the top of a tower is 30° and the angle of depression of the foot of the chimney from the top of the tower is 60°. If the tower is 40 m high, find the height of the chimney.

   Answer:

   $$160/3 \text{ m}$$

   Solution:

   1. Let h be the height of the chimney above the tower level and 40 m be the tower height.
   2. Distance $d = 40\cot(60^\circ) = \frac{40}{\sqrt{3}}$.
   3. Height $h = d\tan(30^\circ) = \frac{40}{\sqrt{3}} \times \frac{1}{\sqrt{3}} = \frac{40}{3}$.
   4. Total height $= 40 + \frac{40}{3} = \frac{160}{3}$ m.

   Derivation:

   $$H = 40 + \frac{40}{\sqrt{3}} \cdot \frac{1}{\sqrt{3}} = 40 + \frac{40}{3} = \frac{160}{3}$$

   Some Applications of Trigonometry — Application with two points

2. **2.** A straight highway leads to the foot of a tower. A man standing at the top of the tower observes a car at an angle of depression of 30°, which is approaching the foot of the tower with a uniform speed. Six seconds later, the angle of depression of the car is found to be 60°. Find the time taken by the car to reach the foot of the tower from this point.

   Answer:

   $$3\text{seconds}$$

   Solution:

   1. Let h be the height of the tower and v be the uniform speed of the car.
   2. Let A be the top of the tower and BC be the distance covered by the car in 6 seconds.
   3. In $\triangle ABC$ (with angle 60° at the car), $\tan 60^\circ = h/CD$.
   4. In $\triangle ABD$ (with angle 30° at the car), $\tan 30^\circ = h/BD$.
   5. Relate the distances using the speed and time formula $d = v \times t$.
   6. Solve for the time t taken to travel the remaining distance CD.

Derivation: Let h be height. $\tan 60° = h/x \implies x = h/\sqrt{3}$. $\tan 30° = h/(x+d) \implies x + d = h\sqrt{3}$. Substituting x : d = h$\sqrt{3} - h/\sqrt{3} = 2h/\sqrt{3}$. Since distance d is covered in 6s, speed v = d/6 = h/(3$\sqrt{3}$) . Time for distance x is t = x/v = (h/$\sqrt{3}$)/($h/3\sqrt{3}$) = 3 seconds. *Some Applications of Trigonometry — Angle of depression*

3. **3.** The angle of elevation of the top of a tower from a point on the ground is 30°. On walking 40 metres towards the tower, the angle of elevation changes to 60°. Find the height of the tower and the distance of the initial point from the tower.

   Answer:

$$\text{Height} = 20\sqrt{3}m, \text{Distance} = 20\sqrt{3} + 40m$$

   Solution:

   1. Let the height of the tower be h and the distance from the second point be x.
   2. Set up the equation for the first point: $\tan 30° = h/(x + 40)$.
   3. Set up the equation for the second point: $\tan 60° = h/x$.
   4. Express h in terms of x from the second equation.
   5. Substitute into the first equation to solve for x and then h.

   Derivation: 1) h/x = $\sqrt{3} \implies h = x\sqrt{3}$. 2) h/(x+40) = $1/\sqrt{3} \implies x\sqrt{3}/(x+40) = 1/\sqrt{3} \implies 3x = x + 40 \implies 2x = 40 \implies x = 20$. Height h = $20\sqrt{3}m$. Initial distance = 20 + 40 = 60 m. *Some Applications of Trigonometry — Angle of elevation*

4. **4.** A vertical tower stands on a horizontal plane and is surmounted by a vertical flagstaff of height 6 metres. At a point on the plane, the angle of elevation of the bottom and top of the flagstaff are 30° and 60° respectively. Find the height of the tower.

   Answer:

$$3\text{metres}$$

   Solution:

   1. Let h be the height of the tower and d be the distance from the point to the base.
   2. Write the equation for the bottom of the flagstaff: $\tan 30° = h/d$.
   3. Write the equation for the top of the flagstaff: $\tan 60° = (h + 6)/d$.
   4. Solve the system of equations for h.

   Derivation: d = h/tan 30° = $h\sqrt{3}$. Substituting into second equation: $\sqrt{3} = (h + 6)/(h\sqrt{3}) \implies 3h = h + 6 \implies 2h = 6 \implies h = 3m$. *Some Applications of Trigonometry — Height of tower with flagstaff*

5. 5. From the top of a 7 m high building, the angle of elevation of the top of a cable tower is 60° and the angle of depression of its foot is 45°. Determine the height of the tower.

Answer:

$$7(1+\sqrt{3})\text{metres}$$

Solution:

1. Let the building be $AB = 7$ m and the tower be CD.
2. Draw a horizontal line from A to meet CD at E. Thus $AE = BD$.
3. In $\triangle ABD$, since the angle of depression is 45°, $BD = AB = 7$ m.
4. In $\triangle ACE$, $\tan 60^\circ = CE/AE$.
5. Calculate CE and add $ED = AB = 7$ m to get the total height.

Derivation: AE = BD = 7 . In △ ACE , CE = AE $\tan 60^\circ = 7\sqrt{3}$. Total height CD = CE + ED = $7\sqrt{3} + 7 = 7(\sqrt{3}+1)m$. *Some Applications of Trigonometry — Combined elevation and depression*

6. 6. A tangent PQ at a point P of a circle of radius 5 cm meets a line through the centre O at a point Q so that $OQ = 13$ cm. The length of PQ is:

Answer: (C)

Solution:

1. The radius is perpendicular to the tangent at the point of contact.
2. In the right-angled triangle OPQ, by Pythagoras theorem, $OQ^2 = OP^2 + PQ^2$.
3. Substitute the values: $13^2 = 5^2 + PQ^2$.
4. Solve for PQ.

Derivation:

$$PQ = \sqrt{OQ^2 - OP^2} = \sqrt{13^2 - 5^2} = \sqrt{169 - 25} = \sqrt{144} = 12 \text{ cm}$$

Circles — Tangents to a circle

7. 7. If tangents PA and PB from a point P to a circle with centre O are inclined to each other at an angle of 80°, then $\angle POA$ is equal to:

Answer: (A)

Solution:

1. The line segment joining the centre to the external point bisects the angle between the tangents.
2. Thus, $\angle APO = 80^\circ/2 = 40^\circ$.
3. In $\triangle OAP$, $\angle OAP = 90^\circ$ (tangent is perpendicular to radius).
4. In $\triangle OAP$, $\angle POA = 180^\circ - 90^\circ - 40^\circ$.

Derivation:

$$\angle POA = 180^\circ - 90^\circ - 40^\circ = 50^\circ$$

Circles — Properties of tangents

8. **8.** From a point Q, the length of the tangent to a circle is 24 cm and the distance of Q from the centre is 25 cm. The radius of the circle is:

   Answer: (D)

   Solution:

   1. Let r be the radius. The triangle formed by the radius, tangent, and line to the centre is right-angled.
   2. $r^2 + 24^2 = 25^2$.
   3. $r^2 = 625 - 576 = 49$.

   Derivation:

   $$r = \sqrt{49} = 7 \text{ cm}$$

   Circles — Tangents to a circle

9. **9.** Two concentric circles are of radii 5 cm and 3 cm. The length of the chord of the larger circle which touches the smaller circle is:

   Answer: (B)

   Solution:

   1. Let the chord be AB and the point of contact be M. $OM \perp AB$.
   2. In $\triangle OMA$, $OA = 5$ (hypotenuse) and $OM = 3$ (radius).
   3. $AM = \sqrt{5^2 - 3^2} = 4$.
   4. Since the perpendicular from the centre bisects the chord, $AB = 2 \times AM$.

   Derivation:

   $$AB = 2 \times 4 = 8 \text{ cm}$$

   Circles — Concentric circles

10. **10.** The number of tangents that can be drawn from a point inside a circle is:

    Answer: (C)

    Solution:

    1. A tangent touches the circle at exactly one point.
    2. A point inside the circle lies in the interior region.
    3. Any line passing through an interior point will cut the circle at two points (secant).
    4. Therefore, no tangent can be drawn.

    Derivation: Number of tangents $= 0$ *Circles — Tangents from a point*

11. 11. Assertion (A): The length of a tangent drawn from an external point to a circle is always equal to the radius of the circle. Reason (R): The tangent at any point of a circle is perpendicular to the radius through the point of contact.

Answer: (D)

Solution:

1. Check Assertion: The length of a tangent from an external point is not necessarily equal to the radius; they are independent.
2. Check Reason: This is a standard theorem (Theorem 10.1 of NCERT).
3. Conclusion: Assertion is false, Reason is true.

Derivation:

$$A : \text{False}, R : \text{True}$$

Circles — Tangents to a circle

12. 12. Assertion (A): If the distance between the centres of two circles of radii r_1 and r_2 is d, then they touch each other externally if $d = r_1 + r_2$. Reason (R): The distance between the centres of two circles is the sum of their radii when they touch internally.

Answer: (C)

Solution:

1. Check Assertion: If two circles touch externally, the distance between their centres is indeed the sum of their radii. True.
2. Check Reason: If two circles touch internally, the distance between their centres is $|r_1 - r_2|$. False.

Derivation:

$$A : \text{True}, R : \text{False}$$

Circles — Relative positions of circles

13. 13. Assertion (A): The number of tangents that can be drawn from a point inside a circle is zero. Reason (R): A tangent to a circle intersects the circle at only one point.

Answer: (A)

Solution:

1. Check Assertion: Points inside the circle cannot have tangents drawn to the circle. True.
2. Check Reason: By definition, a tangent touches the circle at exactly one point. True.
3. Conclusion: Reason explains why no tangent can be drawn from an internal point.

Derivation: A: True, R: True, R explains A *Circles — Tangents to a circle*

14. **14.** Assertion (A): The length of the tangent from a point P at a distance of 10 cm from the centre of a circle of radius 6 cm is 8 cm. Reason (R): In a right triangle, the square of the hypotenuse equals the sum of the squares of the other two sides.

Answer: (A)

Solution:

1. Check Assertion: $L = \sqrt{d^2 - r^2} = \sqrt{10^2 - 6^2} = \sqrt{100 - 36} = \sqrt{64} = 8$. True.
2. Check Reason: Pythagoras theorem is used to calculate the tangent length. True.

Derivation:

$$L = \sqrt{10^2 - 6^2} = 8$$

Circles — Tangents to a circle

15. **15.** Assertion (A): Two parallel tangents can be drawn to a circle at the ends of a diameter. Reason (R): Tangents at the ends of a diameter are perpendicular to the diameter.

Answer: (A)

Solution:

1. Check Assertion: Tangents at ends of diameter are parallel. True.
2. Check Reason: Since the radius is perpendicular to the tangent at the point of contact, and the diameter is a straight line, the tangents are perpendicular to the diameter and thus parallel to each other. True.

Derivation: $90\hat{}\circ + 90^\circ = 180^\circ$(Co-interior angles) *Circles — Tangents to a circle*

16. **16.** Two concentric circles are of radii 5 cm and 3 cm. Find the length of the chord of the larger circle which touches the smaller circle.

Answer:

$$8 \text{ cm}$$

Solution:

1. Let O be the center, AB be the chord of the larger circle tangent to the smaller circle at P.
2. Apply Pythagoras theorem in $\triangle OPA$ where $OA = 5, OP = 3$ to find AP, then double it.

Derivation:

$AP = \sqrt{OA^2 - OP^2} = \sqrt{5^2 - 3^2} = \sqrt{25 - 9} = 4$ cm. Chord $AB = 2 \times AP = 2 \times 4 = 8$ cm.

Circles — Tangent to a circle

17. **17.** If a tangent PA drawn from an external point P to a circle with center O makes an angle of $30°$ with the chord AB, where AB is the diameter, find $\angle APB$.

Answer:

$$60°$$

Solution:

1. Use the property that the angle between tangent and chord equals the angle in the alternate segment.
2. In $\triangle PAB$, use the angle sum property or circle properties.

Derivation:

$$\angle ABP = 90°(\text{angle in a semicircle). Given } \angle PAB = 30°. \text{ In } \triangle PAB, \angle APB = 180° - (90° + 30°) = 60°.$$

Circles — Tangents and chords

18. **18.** In the given figure, if $\angle OAB = 40°$, find $\angle ACB$ where O is the center of the circle.

Answer:

$$50°$$

Solution:

1. Identify that $\triangle OAB$ is isosceles with $OA = OB$.
2. Find $\angle AOB$ and use the theorem that the angle at the center is double the angle at the circumference.

Derivation:

$$\angle OBA = \angle OAB = 40°. \angle AOB = 180° - (40° + 40°) = 100°. \angle ACB = \frac{1}{2}\angle AOB = 50°.$$

Circles — Angle subtended by arc

19. **19.** A quadrilateral ABCD is drawn to circumscribe a circle. If $AB = 6$ cm, $BC = 7$ cm and $CD = 4$ cm, find the length of AD.

Answer:

$$3 \text{ cm}$$

Solution:

1. Use the property that sums of opposite sides of a circumscribed quadrilateral are equal.
2. Substitute the given values into $AB + CD = BC + AD$.

Derivation:

$$\begin{aligned} AB + CD &= BC + AD \\ \implies 6 + 4 &= 7 + AD \\ \implies 10 &= 7 + AD \\ \implies AD &= 3 \text{ cm.} \end{aligned}$$

Circles — Tangents from external point

20. **20.** Two tangents TP and TQ are drawn to a circle with center O from an external point T. If $\angle PTQ = 70°$, find $\angle POQ$.

Answer:

$$110°$$

Solution:

1. Use the property that the opposite angles of a quadrilateral formed by two tangents and two radii are supplementary.
2. Subtract the given angle from 180°.

Derivation:

$\angle OPT = \angle OQT = 90°$. In quadrilateral OPTQ, $\angle POQ + \angle PTQ = 180°$.$\angle POQ = 180° - 70° = 110°$.

Circles — Tangent properties

21. **21.** A quadrilateral ABCD is drawn to circumscribe a circle. Prove that $AB + CD = AD + BC$.

Answer:

$$AB + CD = AD + BC$$

Solution:

1. Use the theorem that tangents drawn from an external point to a circle are equal in length.
2. Identify pairs of equal tangents from vertices A, B, C, and D.
3. Add the four equations to group the sides of the quadrilateral.
4. Substitute the grouped segments to show the sum of opposite sides is equal.

Derivation: Let points of contact be P, Q, R, S on AB, BC, CD, DA respectively. AP=AS, BP=BQ, CQ=CR, DR=DS. Adding them: (AP+BP)+(CR+DR) = (AS+DS)+(BQ+CQ) =¿ AB+CD = AD+BC. *Circles — Tangents to a circle*

22. **22.** Two concentric circles are of radii 5 cm and 3 cm. Find the length of the chord of the larger circle which touches the smaller circle.

Answer:

8 cm

Solution:

1. Draw the radius to the point of contact, which is perpendicular to the tangent.
2. Form a right-angled triangle with the radius of the smaller circle, the radius of the larger circle, and half the chord.
3. Apply Pythagoras theorem to find the length of half the chord.
4. Multiply the result by 2 to get the full chord length.

Derivation: r_1=5, r_2=3. Let half-chord be x. x^2 + 3^2 = 5^2 =¿ x^2 = 25 - 9 = 16 =¿ x = 4. Chord length = 2x = 8 cm. *Circles — Tangents to a circle*

23. **23.** If a tangent PA and PB from a point P to a circle with centre O are inclined to each other at an angle of 80°, then find $\angle POA$.

Answer:

50°

Solution:

1. Identify that the line segment OP bisects the angle between the tangents.
2. Calculate the angle $\angle APO$ as half of $\angle APB$.
3. Use the property that the radius is perpendicular to the tangent at the point of contact.
4. Solve for $\angle POA$ in the right-angled triangle $\triangle OAP$.

Derivation:

$$\angle APB = 80^\circ$$
$$\implies \angle APO = 40^\circ . \angle OAP = 90^\circ . \angle POA = 180^\circ - (90^\circ + 40^\circ) = 50^\circ .$$

Circles — Tangents to a circle

24. **24.** Prove that the lengths of tangents drawn from an external point to a circle are equal.

Answer:

Proof

Solution:

1. Construct a circle with center O and external point P.
2. Draw radii OA and OB to the points of contact A and B.
3. Consider the two triangles $\triangle OAP$ and $\triangle OBP$.
4. Prove congruence using RHS criterion and conclude PA = PB by CPCT.

Derivation: In △ OAP and △ OBP: OA=OB (radii), OP=OP (common), ∠ OAP=∠ OBP=90ˆ∘. By RHS, △ OAP ≅ △ OBP. Thus, PA=PB. *Circles — Tangents to a circle*

25. **25.** A tangent PQ at a point P of a circle of radius 5 cm meets a line through the centre O at a point Q so that $OQ = 12$ cm. Find the length PQ.

Answer:

$$\sqrt{119} \text{ cm}$$

Solution:

1. Recognize that the tangent PQ is perpendicular to the radius OP.
2. Identify $\triangle OPQ$ as a right-angled triangle with $\angle OPQ = 90°$.
3. Apply the Pythagoras theorem: $OQ^2 = OP^2 + PQ^2$.
4. Substitute the given values and solve for PQ.

Derivation:

$$\begin{aligned} OP = 5, OQ = 12.12^2 &= 5^2 + PQ^2 \\ \implies 144 &= 25 + PQ^2 \\ \implies PQ^2 &= 119 \\ \implies PQ &= \sqrt{119} \text{ cm.} \end{aligned}$$

Circles — Tangents to a circle

26. **26.** Prove that the lengths of tangents drawn from an external point to a circle are equal.

Answer:

$$PA = PB$$

Solution:

1. Draw a circle with center O and an external point P.
2. Draw tangents PA and PB touching the circle at A and B respectively.
3. Join OA, OB, and OP.
4. Consider triangles OAP and OBP.
5. Show that the triangles are congruent by RHS congruence criterion.
6. Conclude that PA = PB by CPCT.

Derivation: In △ OAP and △ OBP : OA = OB (radii of the same circle). ∠ OAP = ∠ OBP = 90ˆ∘(tangent is perpendicular to radius at point of contact). OP = OP (common side). Therefore, △ OAP ≅ △ OBP by RHS criterion. Hence, PA = PB by CPCT. *Circles — Tangents to a circle*

27. **27.** PQ is a chord of length 8 cm of a circle of radius 5 cm. The tangents at P and Q intersect at a point T. Find the length of TP.

Answer:

$$TP = 20/3cm$$

Solution:

1. Let the center be O. Join OP and OQ.
2. Let OT intersect PQ at R. Since OP=OQ, $\triangle OPQ$ is isosceles.
3. OT is the perpendicular bisector of PQ, so PR = 4 cm.
4. In $\triangle OPR$, $OR = \sqrt{OP^2 - PR^2} = \sqrt{5^2 - 4^2} = 3$ cm.
5. In $\triangle OPT$, $\angle OPT = 90°$. Using similar triangles $\triangle PRT \sim \triangle OPT$.
6. Set up the ratio: $TP/OP = PR/OR$. Solve for TP.

Derivation: In △ OPT , PR ⊥ OT . Thus, △ PRT ∼ △ OPT (AA similarity). Therefore, TP/OP = PR/OR . TP/5 = 4/3 . TP = 20/3 cm . *Circles — Properties of tangents*

28. **28.** A quadrilateral ABCD is drawn to circumscribe a circle. Prove that $AB+CD = AD + BC$.

Answer:

$$AB + CD = AD + BC$$

Solution:

1. Use the property that tangents from an external point to a circle are equal.
2. Let the points of contact be P, Q, R, S on sides AB, BC, CD, DA respectively.
3. $AP = AS, BP = BQ, CQ = CR, DR = DS.$
4. Sum the left and right sides of these equations.
5. Group the terms to form the sides of the quadrilateral.
6. Conclude the result.

Derivation:

$$AP = AS, BP = BQ, CR = CQ, DR = DS. AB + CD = (AP + PB) + (CR + RD) = (AS + BQ) + (CQ + DS) = (AS + DS) + (BQ + CQ) = AD + BC.$$

Circles — Circumscribed quadrilaterals

29. **29.** Two concentric circles are of radii 5 cm and 3 cm. Find the length of the chord of the larger circle which touches the smaller circle.

Answer:

$$8cm$$

Solution:

1. Let O be the common center. Let AB be the chord of the larger circle touching the smaller circle at point P.
2. Join OP. Since AB is a tangent to the smaller circle, OP $\perp$ AB.
3. OP = 3 cm (radius of smaller circle). OA = 5 cm (radius of larger circle).
4. In right $\triangle OPA$, use Pythagoras theorem to find AP.
5. AP = $\sqrt{OA^2 - OP^2} = \sqrt{5^2 - 3^2} = 4$ cm.
6. Since the perpendicular from the center bisects the chord, AB = 2 * AP.

Derivation: In $\triangle$ OPA , OA^2 = OP^2 + AP^2 $\implies$ $5^2 = 3^2+$ AP^2 $\implies$ $25 = 9+$ AP^2 $\implies$ AP^2 = 16 $\implies$ AP = 4 cm . Since OP $\perp$ AB , AP = PB = 4 cm . Therefore, AB = 4 + 4 = 8 cm . *Circles — Concentric circles*

30. **30.** If the circumference of a circle is 176 cm, then its radius is:

Answer: (C)

Solution:

1. Use the formula for circumference: $C = 2\pi r$
2. Substitute $C = 176$ and $\pi = \frac{22}{7}$
3. Solve for r

Derivation:

$$\begin{aligned} 176 &= 2 \times \frac{22}{7} \times r \\ \implies 176 &= \frac{44}{7} \times r \\ \implies r = \frac{176 \times 7}{44} &= 4 \times 7 = 28 \text{ cm} \end{aligned}$$

Areas Related to Circles — Circumference of a Circle

31. **31.** The area of a circle whose diameter is 14 cm is:

Answer: (B)

Solution:

1. Find the radius: $r = \frac{d}{2} = \frac{14}{2} = 7$ cm
2. Use the area formula: $A = \pi r^2$
3. Calculate A

Derivation:

$$A = \frac{22}{7} \times 7^2 = \frac{22}{7} \times 49 = 22 \times 7 = 154 \text{ cm}^2$$

Areas Related to Circles — Area of a Circle

32. **32.** The area of a quadrant of a circle with radius 7 cm is:

Answer: (A)

Solution:

1. Use the formula for area of a quadrant: $A = \frac{1}{4}\pi r^2$
2. Substitute $r = 7$
3. Calculate the result

Derivation:

$$A = \frac{1}{4} \times \frac{22}{7} \times 7^2 = \frac{1}{4} \times 154 = 38.5 \text{ cm}^2$$

Areas Related to Circles — Area of a Sector

33. **33.** If the perimeter of a semi-circular protractor is 36 cm, then its diameter is:

Answer: (D)

Solution:

1. The perimeter of a semi-circle is $\pi r + 2r = r(\pi + 2)$
2. Set $r(\frac{22}{7} + 2) = 36$
3. Solve for r and find $d = 2r$

Derivation:

$$\begin{aligned} r(\frac{22+14}{7}) &= 36 \\ \implies r(\frac{36}{7}) &= 36 \\ \implies r &= 7 \\ \implies d = 2 \times 7 &= 14 \text{ cm} \end{aligned}$$

Areas Related to Circles — Circumference of a Circle

34. **34.** The area of a sector of a circle with radius 6 cm and central angle 60° is:

Answer: (C)

Solution:

1. Use the formula: $A = \frac{\theta}{360°} \times \pi r^2$
2. Substitute $\theta = 60°$ and $r = 6$
3. Calculate

Derivation:

$$A = \frac{60}{360} \times \pi \times 6^2 = \frac{1}{6} \times \pi \times 36 = 6\pi \text{ cm}^2$$

Areas Related to Circles — Area of a Sector

35. **35.** Assertion (A): The area of a circle with radius 7 cm is 154 cm^2. Reason (R): The area of a circle is given by πr^2, where r is the radius.

Answer: (A)

Solution:

1. Formula for area of a circle is πr^2.
2. Substitute $r = 7$: $\frac{22}{7} \times 7 \times 7 = 22 \times 7 = 154$.

Derivation:

$$\text{Area} = \pi r^2 = \frac{22}{7} \times (7)^2 = 154 \text{ cm}^2$$

Areas Related to Circles — Area of a circle

36. **36.** Assertion (A): The circumference of a circle with diameter 14 cm is 44 cm. Reason (R): The circumference of a circle is given by πd, where d is the diameter.

Answer: (A)

Solution:

1. Circumference formula is πd.
2. Substitute $d = 14$: $\frac{22}{7} \times 14 = 22 \times 2 = 44$.

Derivation:

$$C = \pi d = \frac{22}{7} \times 14 = 44 \text{ cm}$$

Areas Related to Circles — Circumference of a circle

37. **37.** Assertion (A): The area of a semicircle with radius r is $\frac{1}{2}\pi r^2$. Reason (R): The circumference of a semicircle is πr.

Answer: (C)

Solution:

1. Area of semicircle is indeed $\frac{1}{2}\pi r^2$ (True).
2. Circumference of a semicircle includes the diameter, so it is $\pi r + 2r$ (False).

Derivation:

$$\text{Area} = \frac{1}{2}\pi r^2; \text{Perimeter} = \pi r + 2r$$

Areas Related to Circles — Area of semicircle

38. **38.** Assertion (A): If the perimeter and the area of a circle are numerically equal, then the radius of the circle is 2 units. Reason (R): Circumference $2\pi r = \pi r^2$ implies $r = 2$.

Answer: (A)

Solution:

1. Set $2\pi r = \pi r^2$.
2. Divide by πr (since $r \neq 0$), we get $2 = r$.

Derivation:

$$2\pi r = \pi r^2$$
$$\implies r = \frac{2\pi r}{\pi r} = 2$$

Areas Related to Circles — Numerical equality of area and circumference

39. **39.** Assertion (A): The area of a sector of a circle with radius r and central angle θ is $\frac{\theta}{360} \times \pi r^2$. Reason (R): The length of the arc of a sector is $\frac{\theta}{360} \times \pi r^2$.

Answer: (C)

Solution:

1. Area of sector formula is correct (True).
2. Arc length formula is $\frac{\theta}{360} \times 2\pi r$ (False).

Derivation:

$$\text{Area} = \frac{\theta}{360}\pi r^2; \text{Length} = \frac{\theta}{360}2\pi r$$

Areas Related to Circles — Sector of a circle

40. **40.** Find the area of a sector of a circle with radius 6 cm if the angle of the sector is $60°$. (Use $\pi = 3.14$)

Answer:

$$18.84 \text{ cm}^2$$

Solution:

1. Use the formula for the area of a sector: $A = \frac{\theta}{360°} \times \pi r^2$.
2. Substitute the given values $r = 6$ and $\theta = 60$ into the formula and calculate.

Derivation:

$$A = \frac{60}{360} \times 3.14 \times 6^2 = \frac{1}{6} \times 3.14 \times 36 = 3.14 \times 6 = 18.84 \text{ cm}^2$$

Areas Related to Circles — Area of a Sector